Akoizo Original Practice - Form 1

Division C | Mission Possible | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

Name: _____ Date: _____

1. (2 points) How many points is each completed scorable action worth?

Unit: Where the points are: building a scoring plan.
Worked-case data: A Regional device starts on the first coin, completes 8 scorable actions and the Final Action at 58 s, earns all ASL points, sets up in 20 minutes, has no adjustments, and measures 60 x 60 x 70 cm. Estimate its score.
Separate investigation observations:
Plan A: 12 actions, works in 6 of 10 test runs.
Plan B: 9 actions, works in 10 of 10 test runs.
Failure cost: A failed run usually loses the Final Action, 75 no-adjust bonus, and some time points.

A. 50

B. 25

C. 100

D. 250

Answer: _____

2. (2 points) Target Time 90 s. The run takes 100 s. What are the time points?

A. 200

B. 0

C. 160

D. 180

Answer: _____

3. (2 points) What must an endothermic action do to score?

Unit: Chemical actions: gas, cold, and fizz.
Worked-case data: You need about 600 mL of CO_2 to inflate a balloon to size. How much baking soda and vinegar should you use?
Separate investigation observations:
Test 1: 1.0 g baking soda + 30 mL vinegar: balloon expanded 14 cm.
Test 2: 2.0 g + 30 mL vinegar: expanded 21 cm.
Test 3: 3.0 g + 30 mL vinegar: expanded 22 cm; undissolved powder left over.

A. Produce gas

B. Change color

C. Initiate the next action because of the temperature drop

D. Feel cold to the touch

Answer: _____

4. (2 points) When does timing begin?

Unit: The Start Action: coins, slots, and selective triggers.
Worked-case data: Design a rail gap that rejects nickels and dimes but passes quarters to a trigger, allowing 0.5 mm tolerance each side.
Separate investigation observations:
Gap 21.5 mm: Nickel sometimes stuck on the rail and triggered the start.
Gap 22.8 mm: Dime and nickel fell through 20 of 20; quarter carried 20 of 20.
Gap 24.0 mm: Quarter fell through in 6 of 20 trials.

A. When the team signals ready

B. When the gum drops

C. When the first action starts

D. When the coin is inserted

Answer: _____

5. (2 points) A participant pushes the lever that releases the gum. What is the result?

Unit: The Final Action, the ASL, and competition day.
Worked-case data: Estimate the landing speed of a gum pack dropped 55 cm, and suggest how to keep it on the platform.
Separate investigation observations:
Drop tests: Gum fell 52 cm; landed on platform 7 of 10 times, bouncing off 3.
Change: Added a level release shelf; dropped 51 cm.
New tests: Landed and stayed 10 of 10 times.

A. No Final Action points and zero time points

B. A 25-point touch penalty only

C. Full points

D. Tier 3

Answer: _____

6. (4 points) Why should electrical actions be placed early in the sequence?

Unit: Electrical, infrared, and magnetic actions.
Worked-case data: A 9 V battery drives a nichrome segment of 6 Ω. What current and power result, and is the plan legal?
Separate investigation observations:
Timeline: Coin start at 0 s; IR action at 8 s; nichrome cut at 27 s.
Slow test day: Device ran 15 percent slower; cut happened at 31 s.
Supervisor note: Electricity used after 30 s.

Answer: _____
_____
_____

7. (4 points) Why must the incline and pulley masses be removable?

Unit: Simple machines: levers, inclines, pulleys, and screws.
Worked-case data: A screw has a pitch of 1.5 mm. How many turns move an object 2.0 cm, and is that enough rotation for the action?
Separate investigation observations:
Pulley build: Load 420 g lifts 11 cm; effort rope moves 22 cm.
Supervisor check: Counts two supporting segments.
Team claim: Pulley action meets IMA 3.

Answer: _____
_____
_____

8. (4 points) List the four ways to earn ASL points.

Unit: Where the points are: building a scoring plan.
Worked-case data: A Regional device starts on the first coin, completes 8 scorable actions and the Final Action at 58 s, earns all ASL points, sets up in 20 minutes, has no adjustments, and measures 60 x 60 x 70 cm. Estimate its score.
Separate investigation observations:
Plan A: 12 actions, works in 6 of 10 test runs.
Plan B: 9 actions, works in 10 of 10 test runs.
Failure cost: A failed run usually loses the Final Action, 75 no-adjust bonus, and some time points.

Answer: _____
_____
_____

9. (4 points) Why can't a scorable action also be the water timer?

Unit: Timing: Target Time, the water timer, and pacing.
Worked-case data: Target Time 60 s. The device runs 67 s with a 25-s water timer that ended at 40 s. Calculate time points and timer bonus.
Separate investigation observations:
Timer calibration: 150 mL -> 18 s; 300 mL -> 30 s; 450 mL -> 39 s.
Other actions: Total 46 s without the timer.
Target Time: 80 s announced at State.

Answer: _____
_____
_____

10. (4 points) What penalty applies if vinegar splashes out of the device?

Unit: Chemical actions: gas, cold, and fizz.
Worked-case data: You need about 600 mL of CO_2 to inflate a balloon to size. How much baking soda and vinegar should you use?
Separate investigation observations:
Test 1: 1.0 g baking soda + 30 mL vinegar: balloon expanded 14 cm.
Test 2: 2.0 g + 30 mL vinegar: expanded 21 cm.
Test 3: 3.0 g + 30 mL vinegar: expanded 22 cm; undissolved powder left over.

Answer: _____
_____
_____

11. (6 points) Solve this case and explain your reasoning: Design a rail gap that rejects nickels and dimes but passes quarters to a trigger, allowing 0.5 mm tolerance each side.

Answer: _____
_____
_____

12. (6 points) Solve this case and explain your reasoning: Estimate the landing speed of a gum pack dropped 55 cm, and suggest how to keep it on the platform.

Answer: _____
_____
_____

13. (2 points) What change best protects against the penalty?

Supplied synthetic observations:
Timeline: Coin start at 0 s; IR action at 8 s; nichrome cut at 27 s.
Slow test day: Device ran 15 percent slower; cut happened at 31 s.
Supervisor note: Electricity used after 30 s.

A. Move the nichrome cut earlier in the sequence

B. Use a bigger battery

C. Remove the IR action

Answer: _____

14. (2 points) Define “Ideal mechanical advantage (IMA)” in the context of Mission Possible.

Answer: _____
_____
_____

15. (2 points) Define “Scorable action” in the context of Mission Possible.

Answer: _____
_____
_____

16. (2 points) Define “Water timer” in the context of Mission Possible.

Answer: _____
_____
_____

17. (2 points) Define “Nichrome” in the context of Mission Possible.

Answer: _____
_____
_____

18. (2 points) Define “Endothermic” in the context of Mission Possible.

Answer: _____
_____
_____